

Impact of the use of COVID-19 serological data on modeling

Perspective of the SIMID modeling group

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SIMID

Simulation Models of Infectious Diseases

www.simid.be

What can we learn from (serial) serological survey data?

Metcalf et al. (2016)

THE LANCET

Use of serological surveys to generate key insights into the changing global landscape of infectious disease

[Dr C Jessica E Metcalf, PhD](#)   • [Jeremy Farrar, MD](#) • [Prof Felicity T Cutts, MD](#) • [Nicole E Basta, PhD](#) • [Andrea L Graham, PhD](#) • [Justin Lessler, PhD](#) • [Prof Neil M Ferguson, DPhil](#) • [Prof Donald S Burke, MD](#) • [Prof Bryan T Grenfell, PhD](#) • [Show less](#)

Seroprevalence of IgG antibodies against SARS coronavirus 2 in Belgium

A serial prospective cross-sectional nationwide study of residual samples

Aim

- Estimate seroprevalence (and seroincidence) in Belgium

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- **Prospective serial cross-sectional nationwide study**
 - Started end March 2020 (approx. every 3 to 4 weeks)
- **Residual samples**
 - 3000-4000 residual samples per collection period
 - Stratified by age, sex, region
 - No hospital/triage centre samples
 - **Seven collection rounds** (in 2020 – prior to vaccination)
 - 1st to 5th: 30 March 2020 – 05 July 2020
 - 6th: 07 September 2020
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 - “Budget” of 4000 and 3000 samples respectively
 - Focus: precise overall estimate of interest BUT also for relevant subgroups (e.g., age)
 - **Approach**: simulation studies

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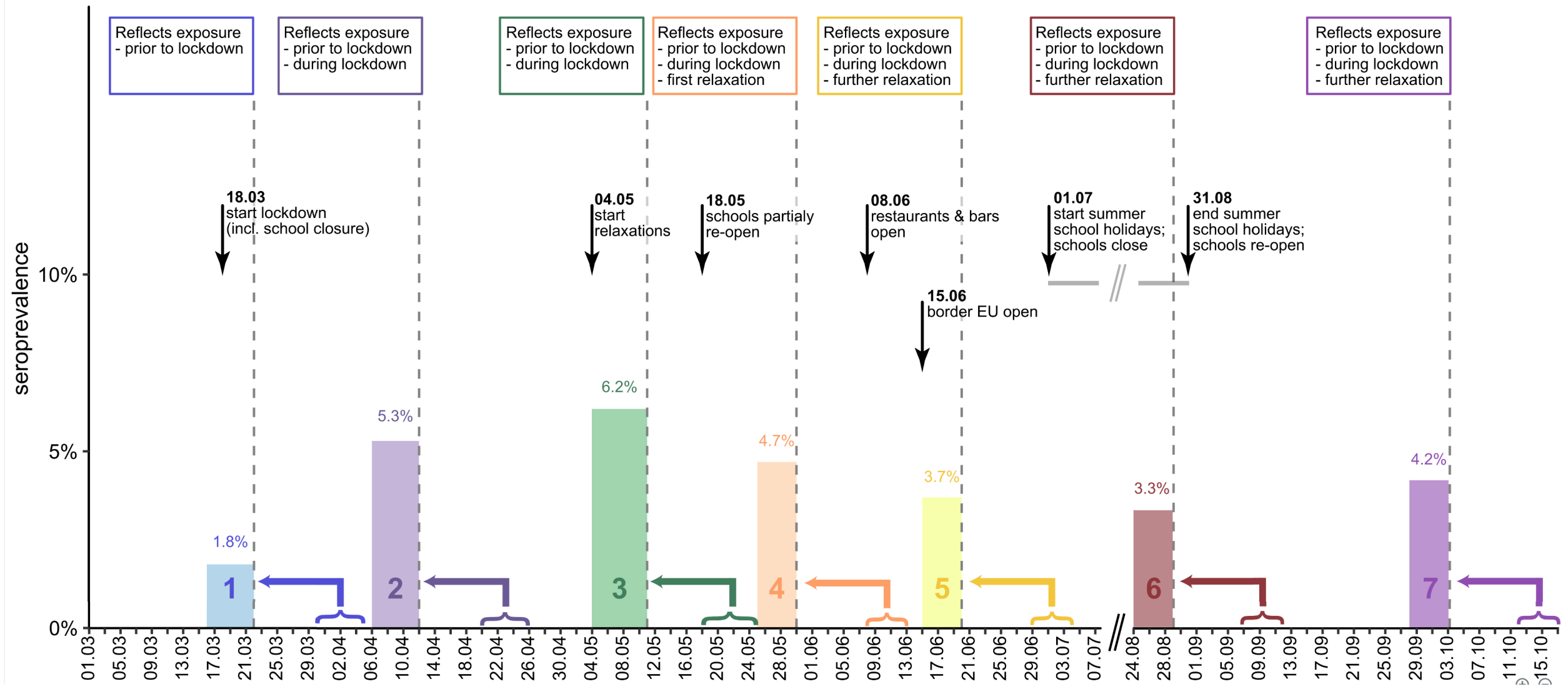
Design and standardisation aspects
related to serological data

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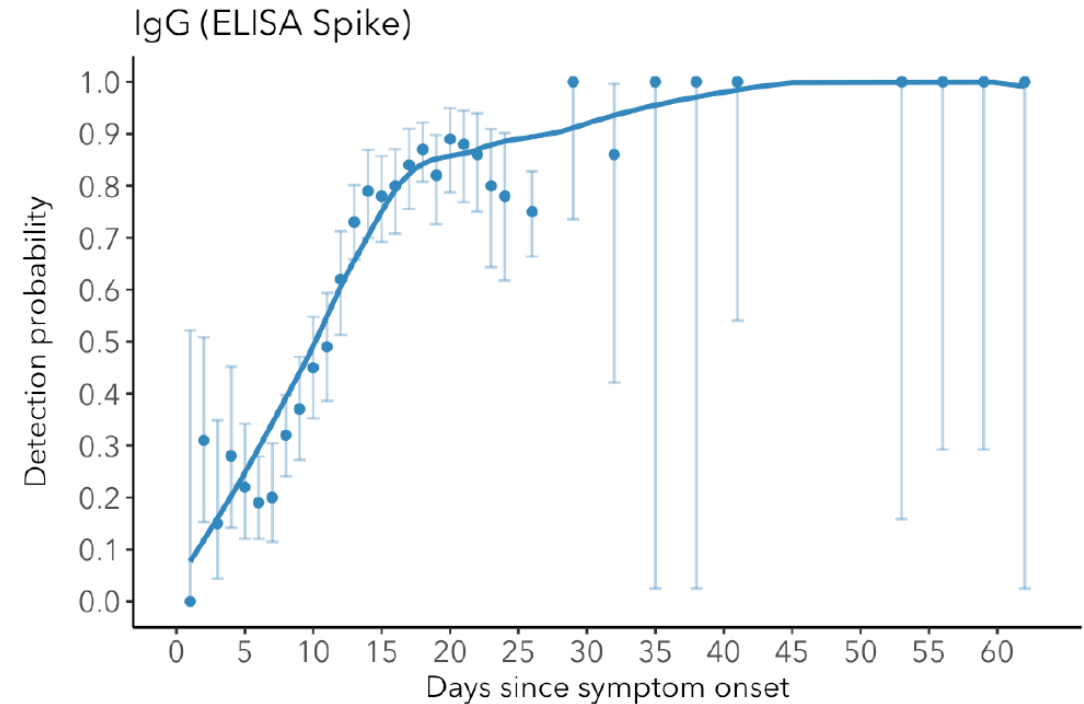
A serial prospective cross-sectional nationwide study of residual samples

Methods

- IgG antibodies against S1 proteins of SARS-CoV-2
 - Semi-quantitative commercial ELISA
 - Sensitivity 81.5% - Specificity 96.3%

Statistical analysis

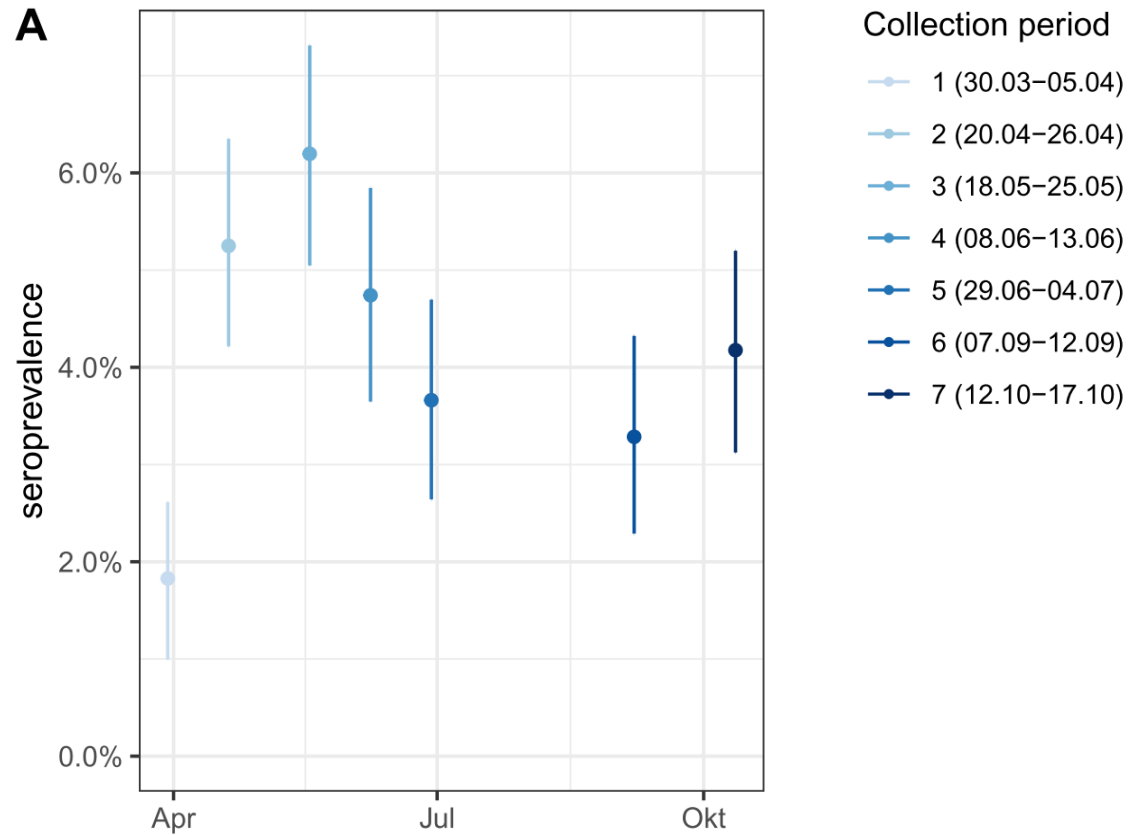
- Serostatus positive if IgG OD values ≥ 1.1
- Bayesian modelling approach
 - Weighted samples
 - Age, sex, provinces
 - Trimmed to max. 3
 - ELISA test sensitivity & specificity (incl. uncertainty)
 - External dataset (study in Geneva, Switzerland)



Borremans et al. (2020) (IgG ELISA Spike - 4 studies)

Seroprevalence of IgG antibodies against SARS coronavirus 2 in Belgium

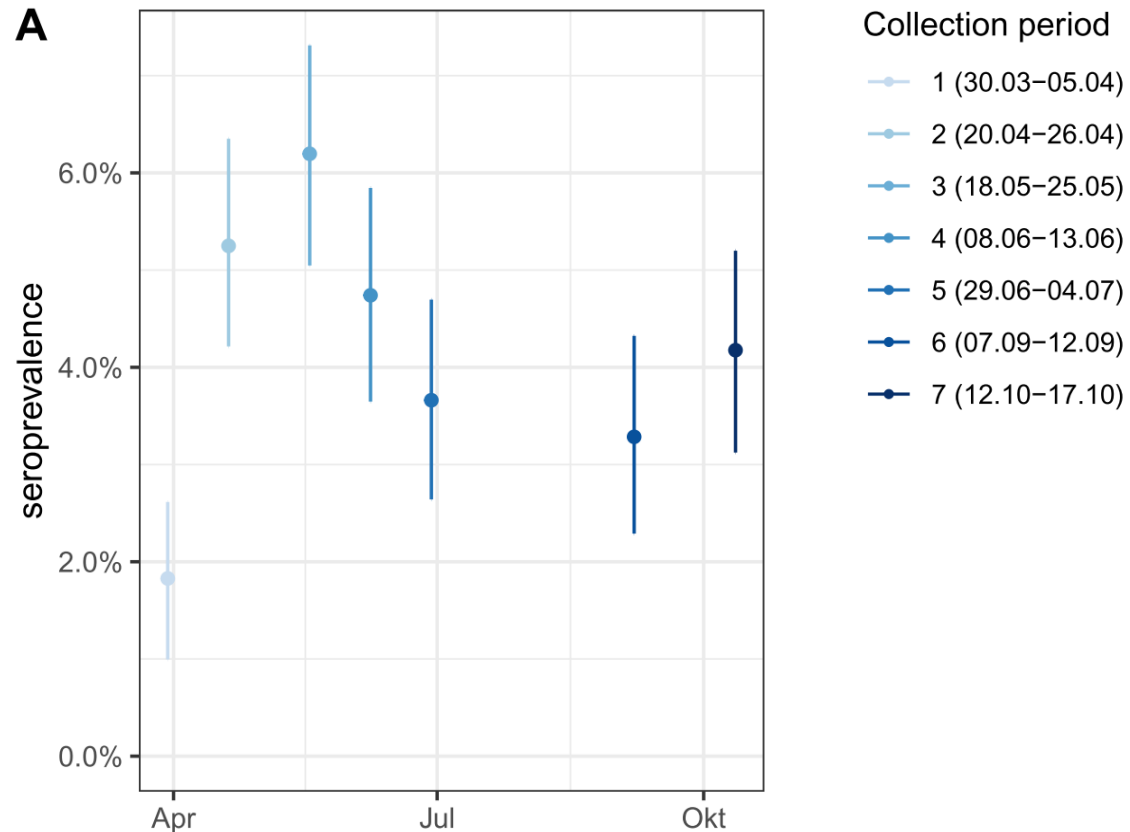
A serial prospective cross-sectional nationwide study of residual samples



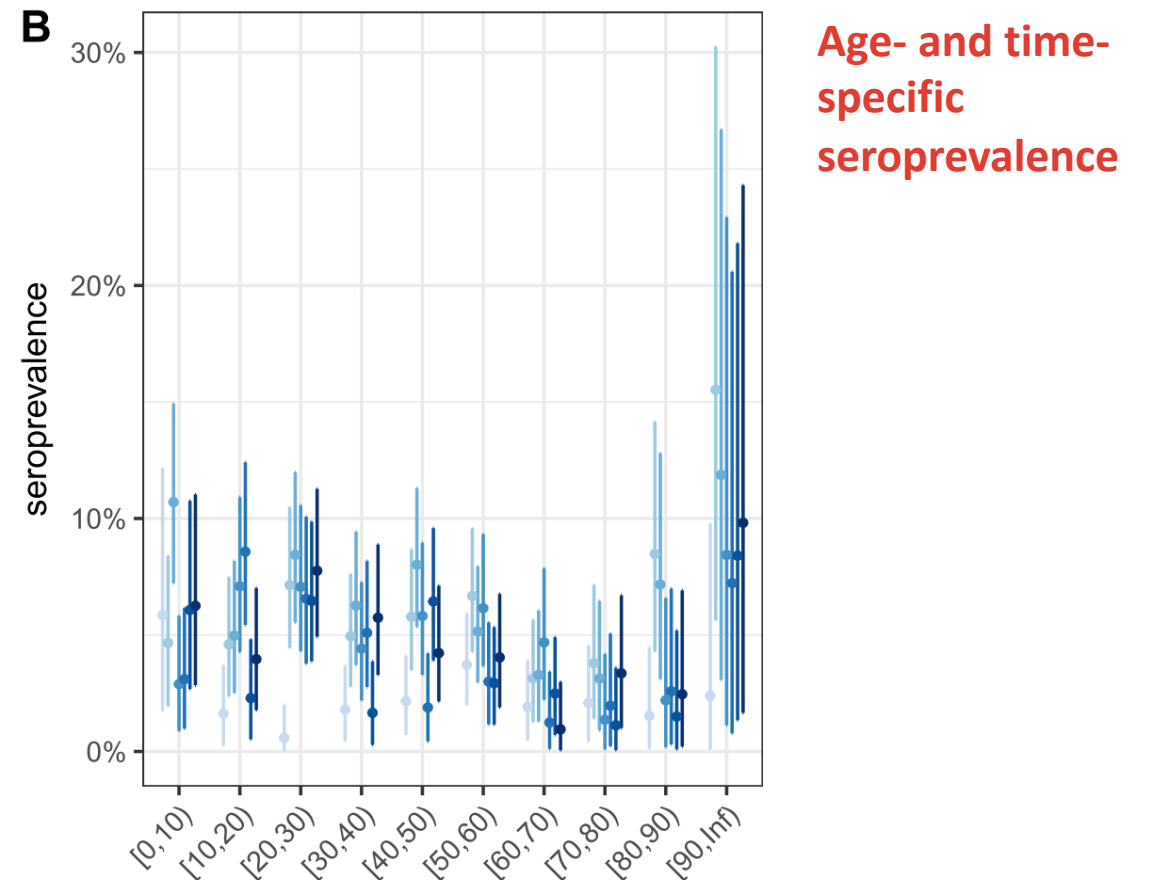
Weighted seroprevalence in Belgium

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Weighted seroprevalence in Belgium



Mathematical modeling of infectious diseases

Serial serological survey data used in SARS-CoV-2 modeling

Purposes

- **Prediction:** requires the inclusion of known complexities and population-level heterogeneity → forecasting
- **Understanding:** investigating the factors that drive dynamics

Model building

- **Accuracy:** reproduce what is observed and predict future dynamics
- **Transparency:** ability to understand how model components influence the dynamics and interact
- **Flexibility:** ease of adapting the model to new situations

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Limitations

- Models present a **simplification of reality**
- **Chance events** of infectious disease transmission **hinder perfect prediction**

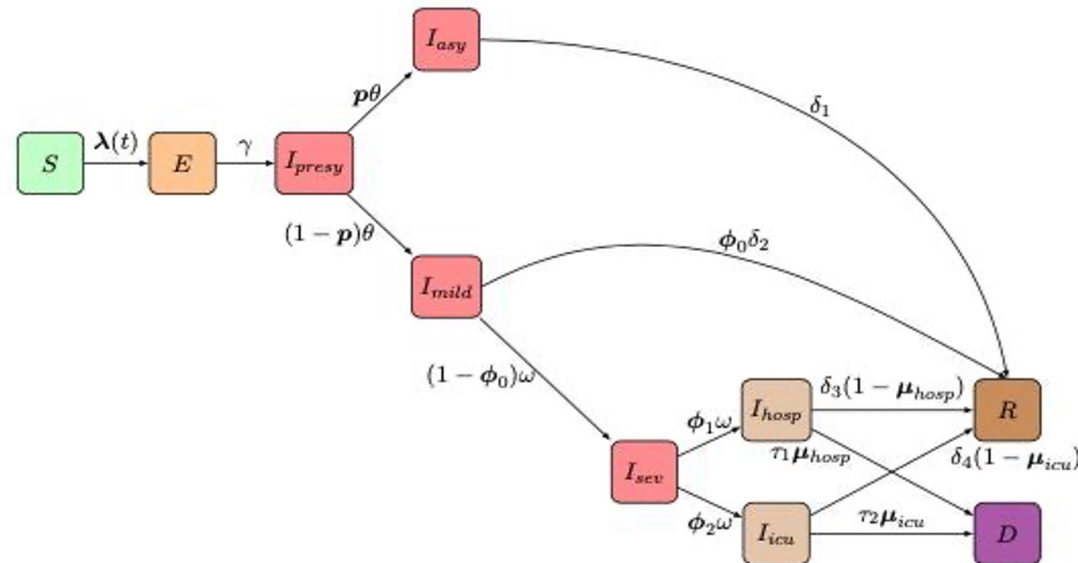
A good model

- Suited to its purpose: **simple as possible, but no simpler**
- **Balance accuracy transparency, flexibility**
- **Parameterisable** from available data **or** alternatively **sensitivity analysis for unknown parameters**

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Schematic diagram of the stochastic model

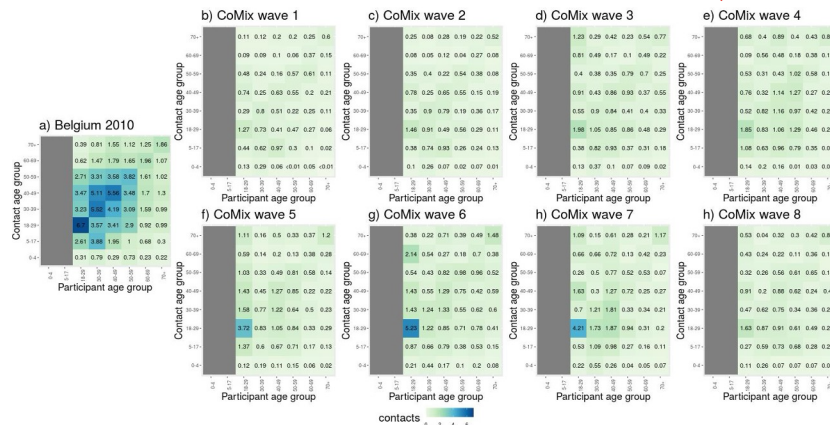


Stochastic discrete-time compartmental model for Belgium

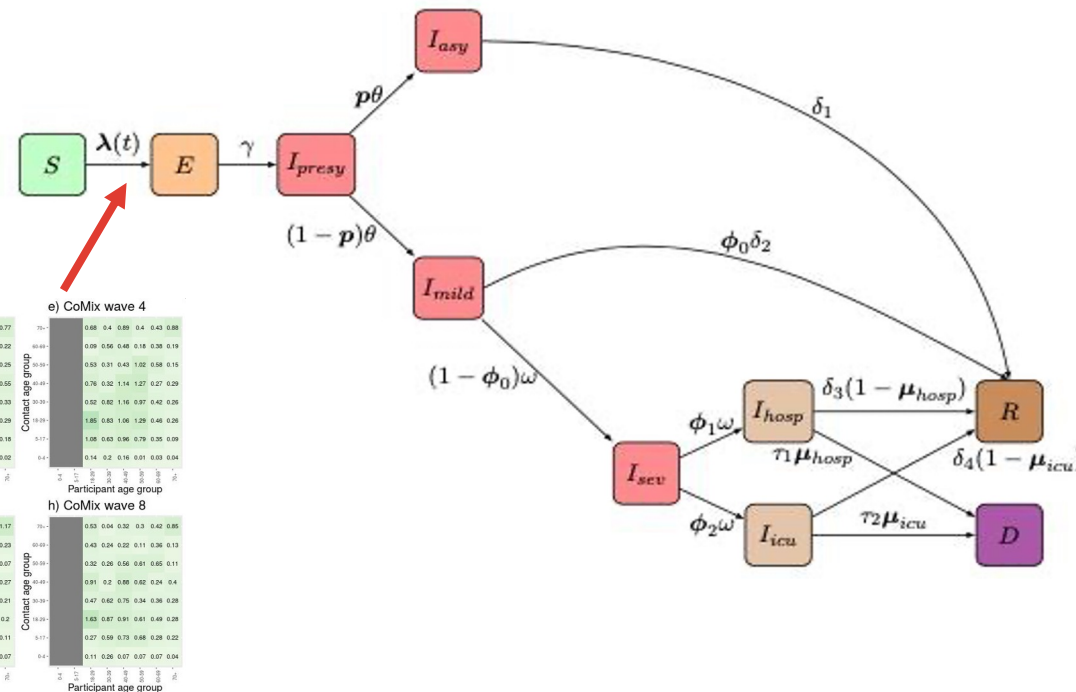
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CoMix data



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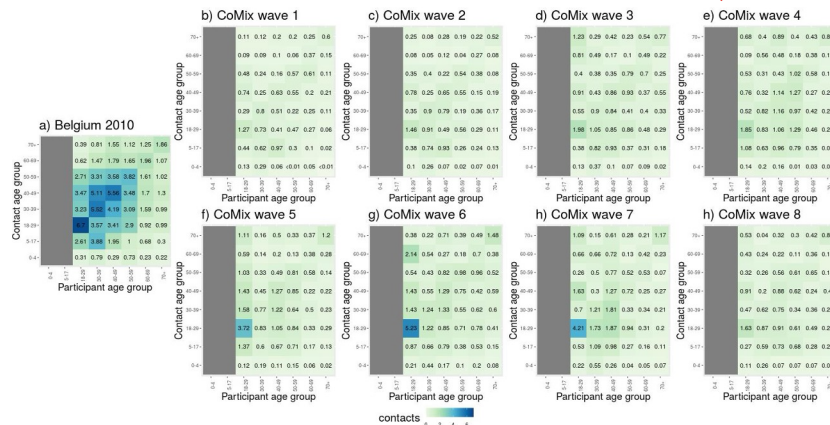
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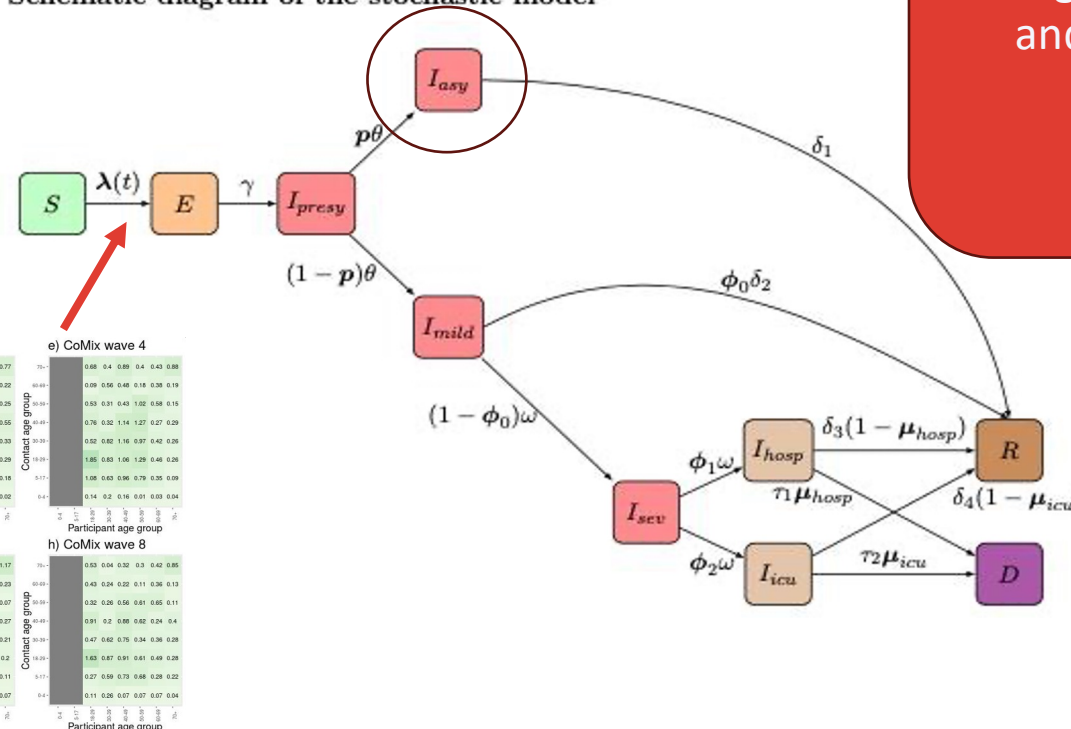
Proportion of asymptomatic individuals

Changes in case definition, temporal and age-specific differences in asymptomatic fraction, underreporting, ...

CoMix data



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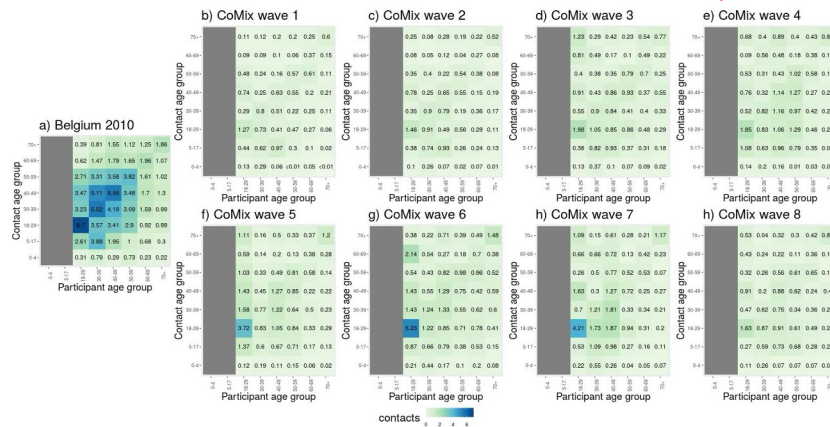


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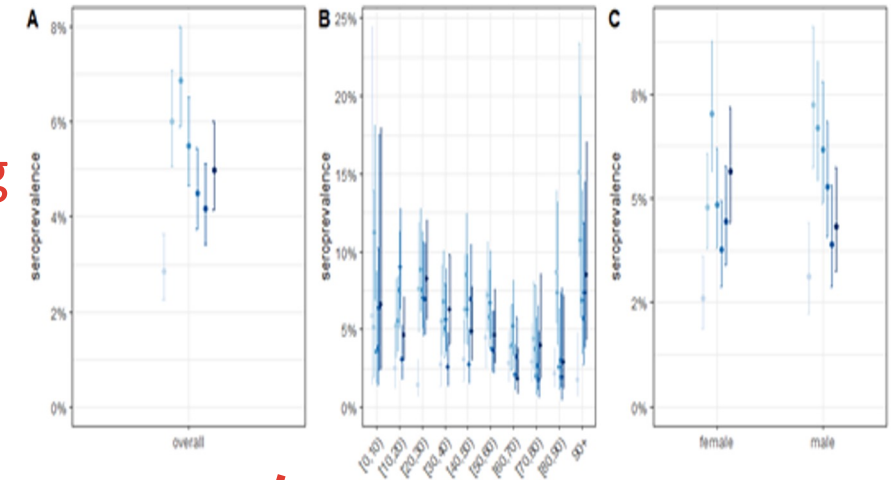
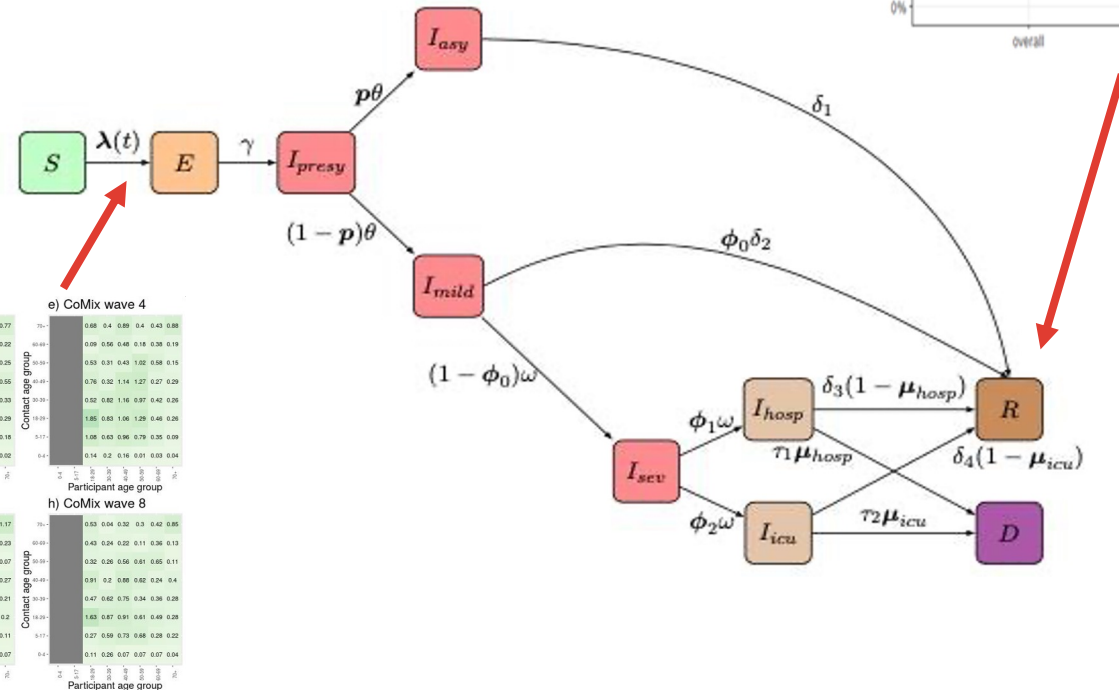
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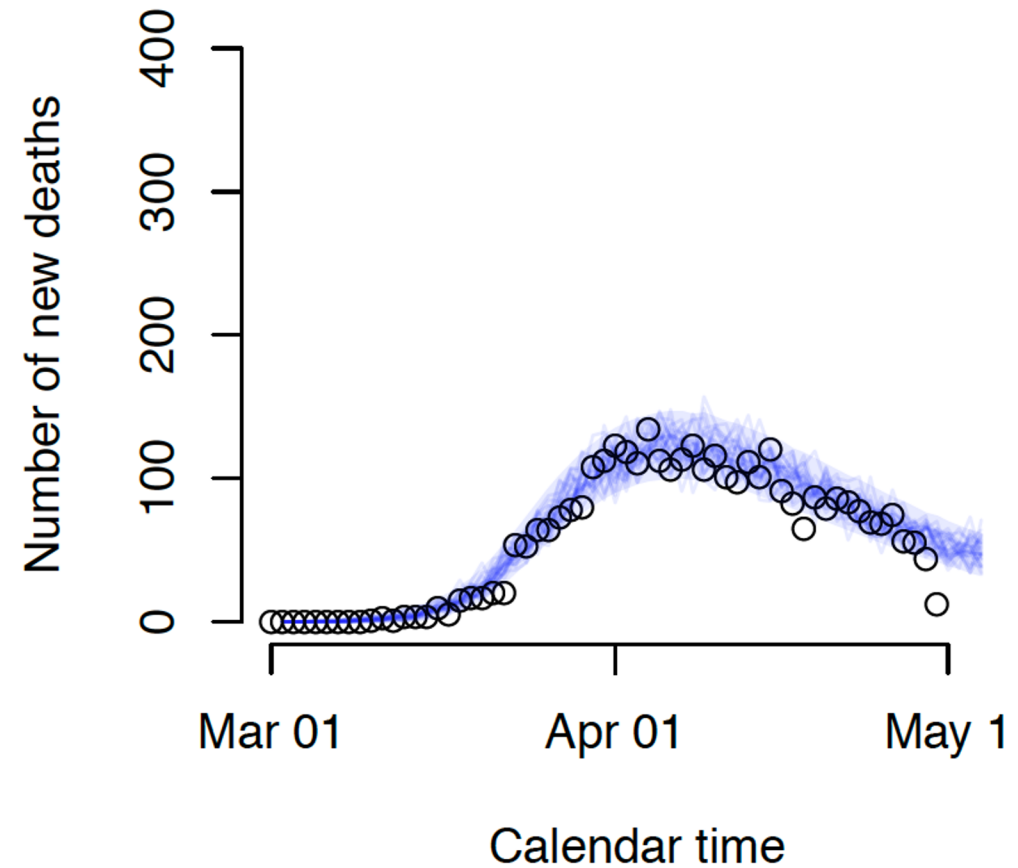
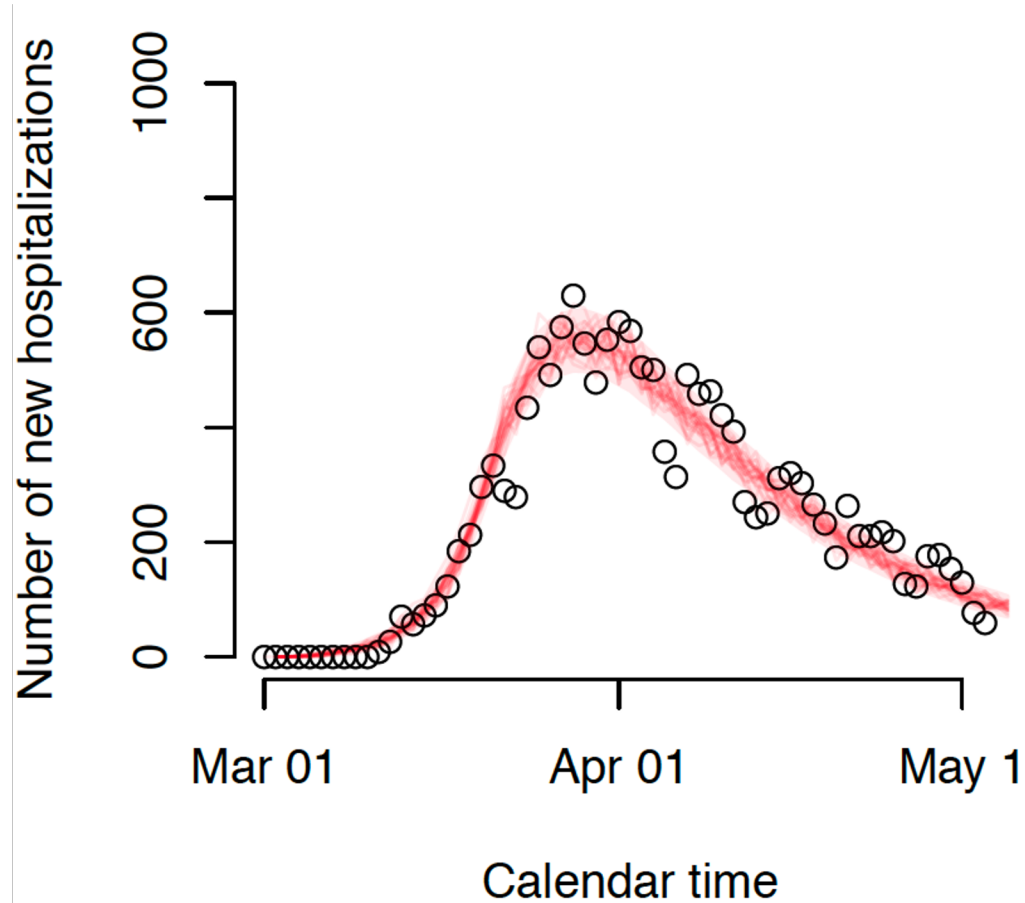
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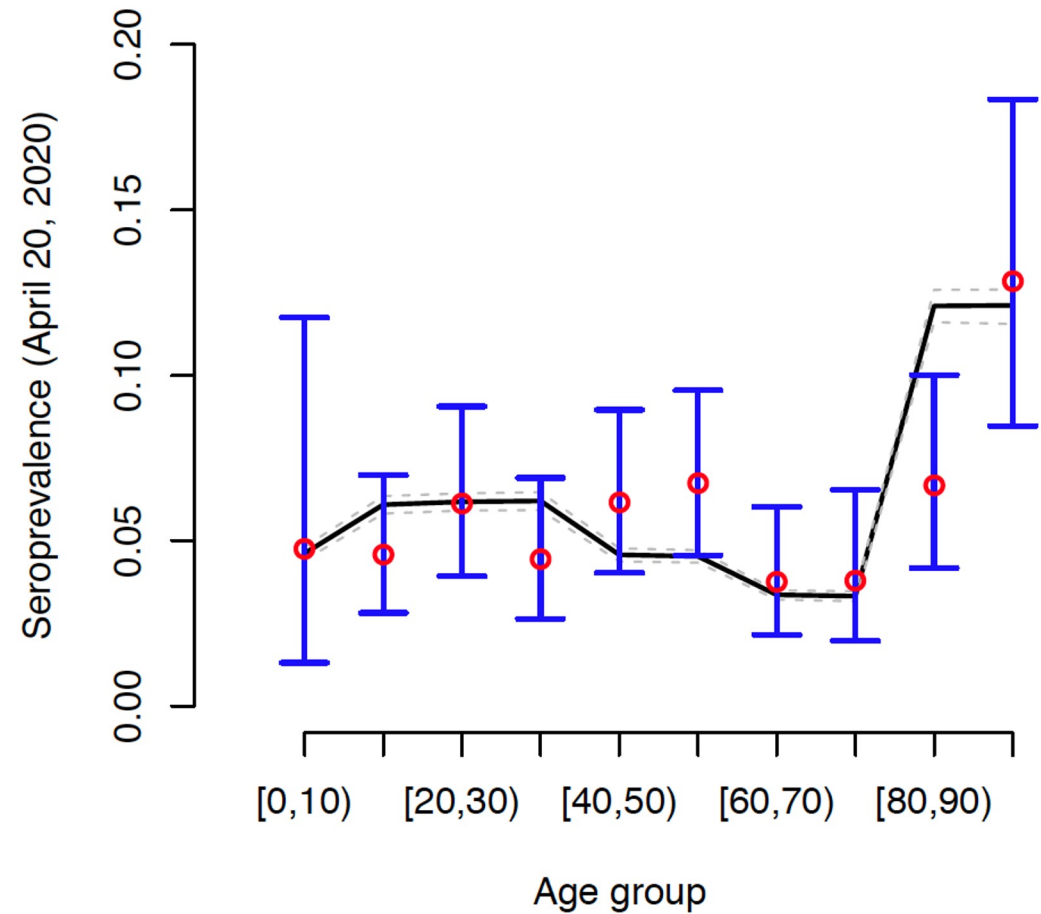
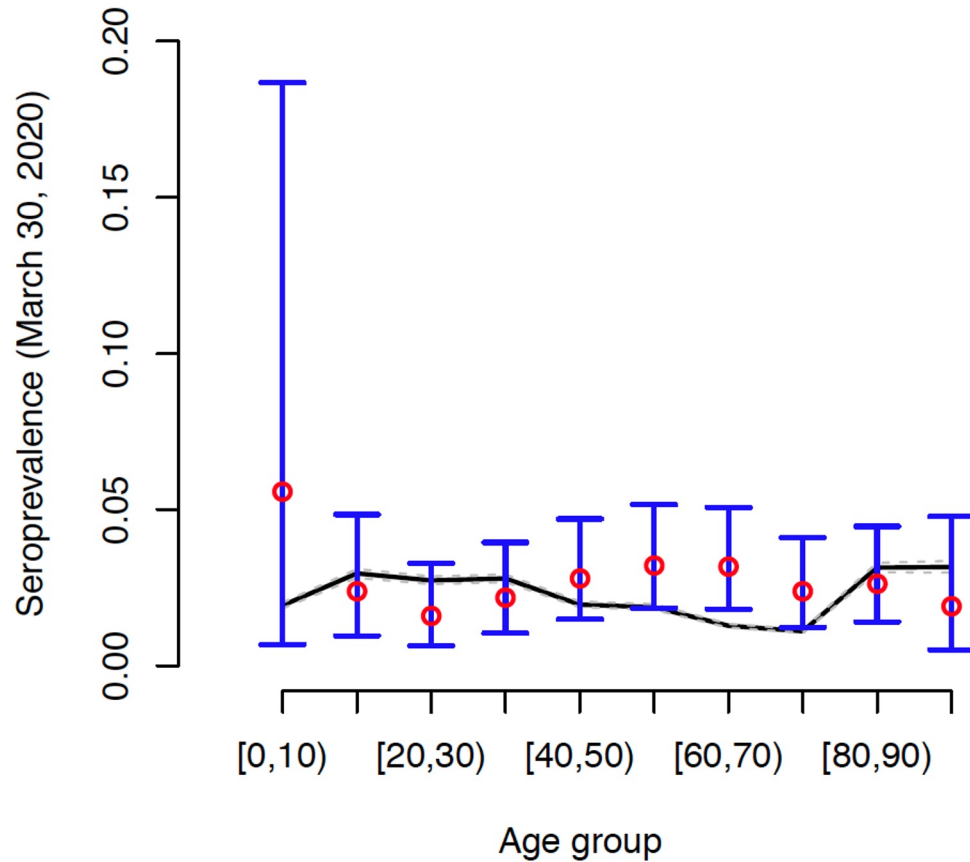
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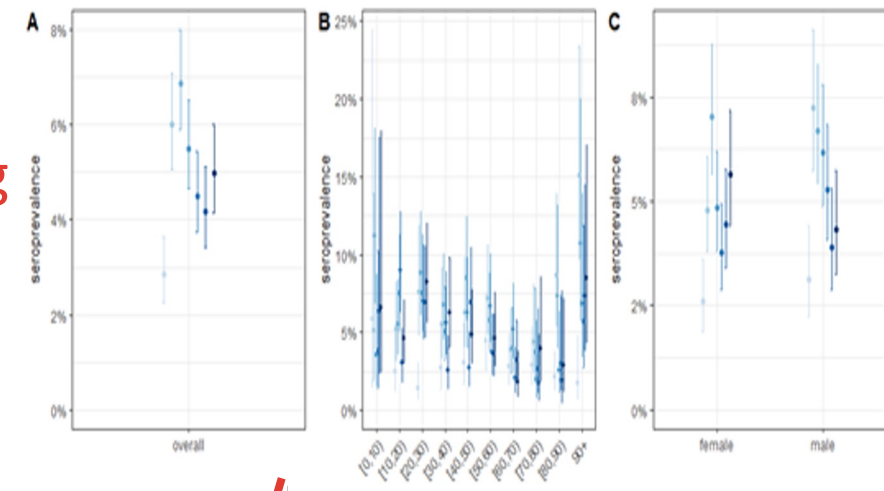
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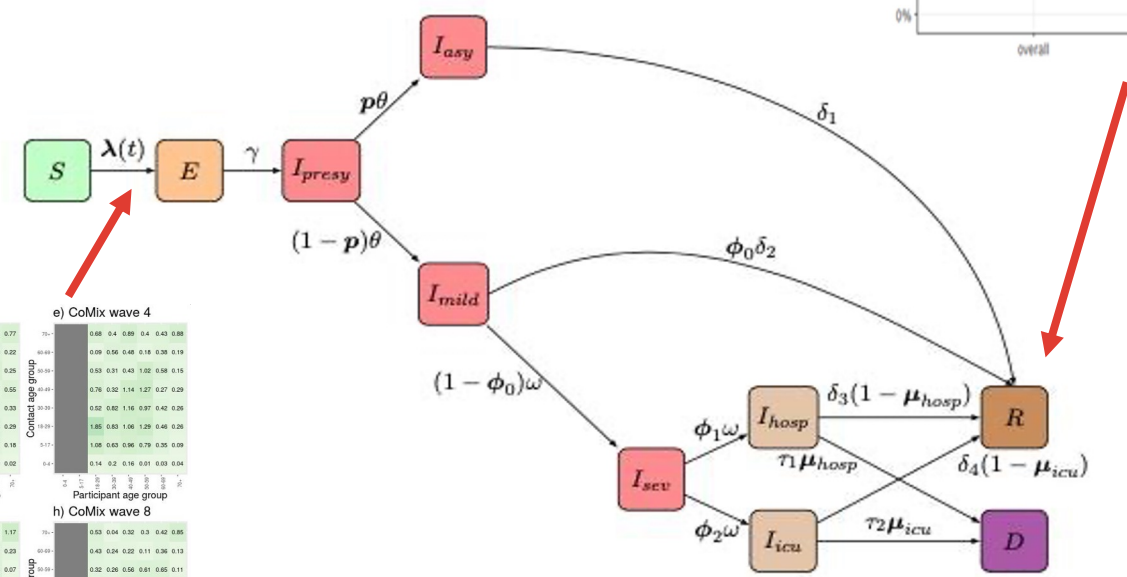


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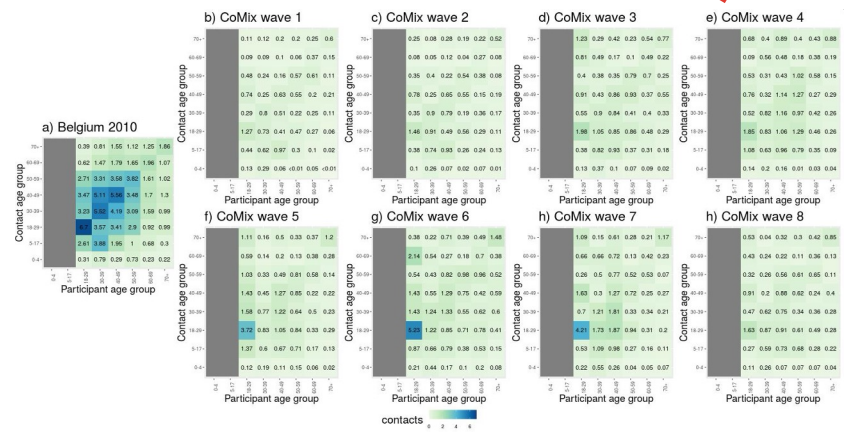
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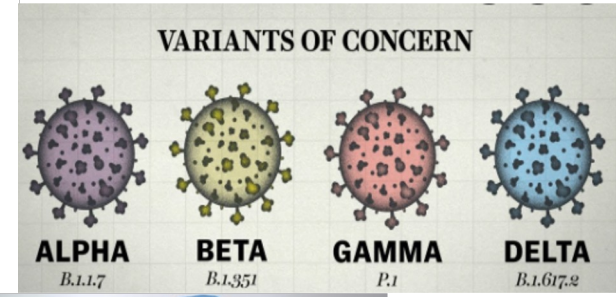
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CoMix data



Extensions:

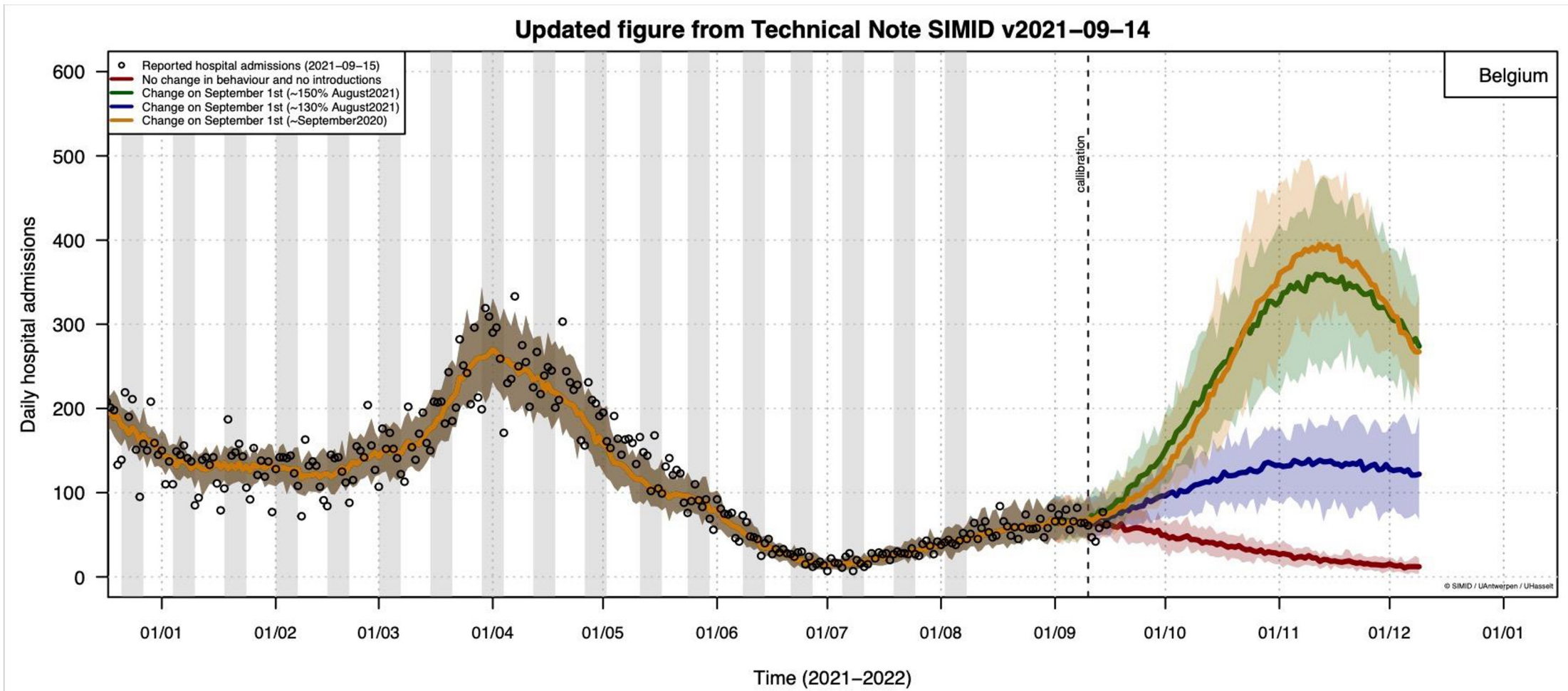


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Reports available at <https://covid-en-wetenschap.github.io/>

Discussion

- The use of **residual samples** (too frail) **versus blood donor data** (too healthy, no children):
 - Need for a random sample of individuals in the population
 - Care-seeking behaviour and induced bias in terms of seroprevalence estimates
- Antibody waning: **longitudinal data to unravel antibody dynamics**
- **Serial serological survey data** of crucial importance to estimate the “dark number”
- **Other modelling work** (some based on the SCM) related to serology:
 - Optimal design of serological studies (SARS-CoV-2, measles, ...)
 - Standardisation of serological survey results with different tests, time periods, etc. (e.g., pertussis)
 - Modeling of continuous antibody titer data (PVB19, VZV, ...)
 - Analysis of paired serological survey data (e.g., frailty modelling)
 - Next-generation approach (e.g., studying the role of children in transmission)
 - ...
- Strengthening **sero-surveillance**, also for **endemic and vaccine-preventable diseases**

Acknowledgements



RESTORE



References

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Thank you



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